MIL-S-12645C
24 February 1984
SUPERSEDING
MIL-S-12645B
9 March 1973

MILITARY SPECIFICATION

SLINGS, ENDLESS, FIBER ROPE AND NYLON WEBBING

'This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers fiber rope and nylon webbing endless slings.

1.2 Classification. Slings shall be the following types, as specified (see 6.2):

Type	Inside perimeter (feet)	Rope circumference (inches)	Nylon webbing width (inches)	Number of ply
I	24	3		
II	40	3-1/4		
III	80	3-1/2		
IV	80	4		
v	24		1	1
VI	40		2	1
VII	80		3	1
VIII	80		4	1

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research and Development Center, ATTN: STRBE-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3940

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2. 1.1 Specifications, and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standard (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

T-R-605	- Rope, Manila and Sisal.
T-T-616	- Treatment: Mildew Resistant, for Rope and Cord.
PPP-B-601	- Boxes, Wood, Cleated Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-640	- Boxes, Fiberboard, Corrugated, Triple Wall.

STANDARDS

FEDERAL

FED-STD-191	- Textile Test Methods.

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of US Military Property.
MIL-STD-1188	Commercial Packaging of Supplies and Equipment.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the test of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 Description. Each sling shall be a continuous length of manila rope or nylon webbing with its end spliced together to form a loop.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.3 Materials. Materials shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.4).

3.3.1 Fiber rope. The fiber rope shall conform to T-R-605, type M, class 2, and shall be mildew resistant treated in accordance with T-T-616.

3.3.2 Nylon webbing. The nylon webbing sling shall be fabricated from continuous-filament heat and light resistant nylon fiber. The nylon shall be a long chain polymer made of hexamethylene diamine and adipic acid, or a long chain polymer of epsilon amino caproic acid. Mixtures of nylon fiber types shall not be employed in any one sling.

3.4 Construction.

3.4.1 Sling charactacristics. Fiber rope slings shall conform to table I and the nylon webbing slings shall conform to table II.

TABLE I. Fiber rope sling characteristics.

Type	Breaking strength (minimum)	Safe working load 1/	Rope circumference (nominal)	Perimeter and tolerance
	pounds	pounds	inches	fee t
I	8,100	1,620	3	24 + 1.5
II	9,450	1,890	3-1/4	40 + 2.0
III	10,800	2,160	3-1/2	80 + 4.0
IV	13,500	2,700	4	80 + 4.0

1/ Safe working load is the maximum recommended load based on a safety factor of 5.

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TABLE II. Nylon webbing sling characteristics.

Type	Breaking strength (minimum)	Safe working load 1/	Sling width	Number of ply	Perimeter and tolerance
	pounds	pounds	inches		feet
V	9,000	1,800	1	1	24 ± 1.5
VI	15,000	3,000	2	1	40 ± 2.0
VII	22,500	4,500	3	1	80 ± 4.0
VIII	30,000	6,000	4	1	80 ± 4.0

1/ Safe working load is the maximum recommended load based on a safety factor of 5.

3.4.2 Splices. Fiber rope short splices shall have five tucks on each side, consisting of three fullstrand tucks and two half-strand tucks. Before the last two tucks of each strand are made, the strands shall be reduced in thickness by one-half to taper the splice. Each tuck shall be pulled up snugly to form 3 tight and neat splice. The ends of each strand shall protrude at least 3/4 inch beyond the last tuck. Nylon webbing shall be spliced by stitching with nylon thread.

3.5 Performance. The sling shall meet the breaking strength requirements shown in tables I and 11 when subjected to the test specified in 4.5.2.1.

3.6 Identification. Identification shall be in accordance with MIL-STD-130. A metal tag shall be securely attached to each sling. Metal tags shall be corrosion resistant and of 0.016 inch minimum thickness, securely fastened with corrosion resistant wire of 0.047 inch minimum diameter. This tag shall have the following information etched or embossed legibly thereon:

1.	Nomenclature.	Sling, endless (fiber rope or nylon webbing)
2.	Specification and type.	MIL-S-12645
3.	National stock number.	NSN 3940-_____
4.	Safe working load, pounds (see 3.4.1).	Safe working load: _____ pounds
5.	Manufacturer's code and Contract No.	MFR's Code _____ : Contract No. _____
6.	U.S.	U.S.

3.7 Workmanship. The finished slings shall be clean and shall conform to the quality and grade of product as required by this specification.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspections. Inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article sling(s) shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Test. The first article sling(s) shall be tested as specified in 4.5.2.1. Failure of the test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling.

4.4.1.1 Examination. Sampling for examination shall be in accordance with MIL-STD-105.

4.4.1.2 Test. Sampling for test shall be in accordance with MIL-STD-105, inspection level S-2.

4.4.2 Examination. Samples selected in accordance with 4.4.1.1 shall be examined for the defects specified in 4.5.1. The AQL shall be 1.5 percent defective.

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4.4.3 Test. Samples selected in accordance with 4.4.1.2 shall be tested in accordance with 4.5.2.1. Failure of any sample shall be cause for rejection of the entire lot represented by the sample.

4.5 Inspection Procedure.

4.5.1 Examination. The sling shall be examined as specified herein for the following defects:

101. Materials not as specified.
102. Sling does not meet dimensional requirements.
103. Sling not spliced as specified.
104. Identification marking missing, incorrect or illegible.
105. Workmanship not as specified.

4.5.2 Test.

4.5.2.1 Breaking strength. The sling shall be tested for breaking strength in accordance with FED-STD-191, method 6015, except that the tensile load shall be applied through mandrels not less than 4 inches in diameter. The specimen of the sling to be tested shall be a single length of sling which shall be cut from that portion of the sling containing the splice and which shall be positioned so that the splice will be midway between the mandrels when making the tensile strength tests. Nonconformance to 3.5 shall constitute failure of the test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

106. Containers not as specified for level A or B.
107. Box closure and sealing not as specified for level A.
108. Slings of unlike description packed together for level A or B.
109. Quantities of slings packed together exceed the size or weight limitations of the applicable box specification for level A or B.
110. Preservation and packing not in accordance with the referenced document for commercial.
111. Marking missing, illegible, incorrect or incomplete.

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5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

5.1.1 Level A. Each sling shall be coiled to the minimum practicable size, secured to prevent uncoiling and placed in a close-fitting box-conforming to PPP-B-636, type CF, class weather resistant, variety SW, grade as applicable to the weight of the sling. Box closure and sealing shall be as specified for Method V in accordance with the appendix to the box specification.

5.1.2 Commercial. Slings shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be level A, B or commercial as specified (see 6.2).

5.2.1 Level A. Slings of like description shall be packed together in a close-fitting box conforming to PPP-B-601, overseas type, style optional, in quantities not to exceed the weight limitation of the box. Box closure and strapping shall be in accordance with the appendix to the box specification.

5.2.2 Level B. Slings of like description. shall be packed together in a close-fitting box conforming to PPP-B-601 domestic type, style optional; or in a close-fitting fiberboard box conforming to PPP-B-640, class 2, style optional, or PPP-B-636, V3c Vllc, V13c or V15c. The gross weight or size of each box shall not exceed the limitations of the applicable box specification. Box closure and strapping shall be in accordance with the appendix to the applicable box specification.

5.2.3 Commercial. Slings shall be packed in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military. Marking for military levels of protection (level A or B) shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Marking for commercial packaging shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The endless slings are intended for use in cargo and material handling operations.

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6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Type of sling required (see 1.2).

c. When a first article is required for inspection and approval and the number of units required (see 3.2, 4.3 and 6.3).

d. Degree of preservation and degree of packing required (see 5.1 and 5.2).

6.3 First article. When a first article inspection is required, the item will be tested and should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, test and approval of the first article.

6.4 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.3).

Custodians:
Army - ME
Air Force - 99

Review activities:
Army - MT, AV
Air Force - 84
DLA - IS

User Activity:
Navy - YD, SH

Preparing activity:
Army - ME

Project 3940-0164

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-12645C	Slings, Endless, Fiber Rope and Nylon Webbing

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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